

Carnegie Research Workshop

Continental Climate Change: Simple Models to Understand the Future

6th – 8th June 2022, University of St Andrews

Organisers

- Mike Byrne (University of St Andrews and University of Oxford)
- Gabi Hegerl (University of Edinburgh)
- Jack Scheff (University of North Carolina at Charlotte)

Format of talks: Each speaker is allocated a 30-minute slot. In order to leave plenty of time for questions/discussion, please aim to speak for no more than 23 minutes.

Programme

Sunday, 5th June

- 3pm onwards: Attendees check-in at Agnes Blackadder Hall
- 3.30pm: Optional hike along West Sands (approx. 4 miles, easy terrain). Meet at entrance to ABH!
- 6pm: Informal welcome meetup at [St Andrews Brewing Co.](#)
- 7pm-: Dinner on your own

Monday, 6th June

- 9-9.20am: Registration (School 5)
- 9.20-9.30am: Welcome comments, logistics, etc. (Mike Byrne, Gabi Hegerl, Jack Scheff)
- **Session 1, Part 1: Fundamental Role of Land in the Climate System** (Chair: Jack Scheff)
 - 9.30-10am **Marysa Laguë** – Influence of continental configuration on the base-state of atmospheric water vapour
 - 10-10.30am **Hugo Lambert** – A simple model of land precipitation over an idealised tropical continent
- 10.30-11am: COFFEE BREAK (Lower College Hall)
- **Session 1, Part 2: Fundamental Role of Land in the Climate System** (Chair: Marianne Pietschnig)
 - 11-11.30am **Luca Schmidt** – OCELAND: A Conceptual Model to Explain the Partitioning of Precipitation between Ocean and Land
 - 11.30-12pm **Cathy Hohenegger** – Do tropical continents receive more than their fair share of precipitation?
 - 12-12.30pm **Matthew Henry** – Reduced High-Latitude Land Seasonality in Climates with Very High Carbon Dioxide.
- 12.30-1.30pm: LUNCH (Lower College Hall)
- **Session 1, Part 3: Fundamental Role of Land in the Climate System** (Chair: Matthew Henry)
 - 1.30-2pm **Axel Kleidon** – Doing climate science on the back of an envelope: Using thermodynamic limits to infer continental climate change
 - 2-2.30pm **Ori Adam** – Moisture constraints on tropical seasonal and diurnal temperature variations
- 2.30-3pm: DISCUSSION on Session 1 (Moderator: Kaighin McColl)
- 3-3.30pm: COFFEE BREAK (Lower College Hall)
- **Session 2, Part 1: Land, Monsoons and Precipitation** (Chair: Ruth Geen)

- 3.30-4pm **Simona Bordoni** – Changes in the timing of monsoon onset with warming in the model hierarchy
- 4-4.30pm **Michela Biasutti** – Ventilation revisited: how dry continental air splits the ITCZ and stops the monsoon
- 4.30-5pm **Sonia Seneviratne** – Drought-heatwave feedbacks in a changing climate (*virtual presentation*)
- 5-5.30pm: **Special talk by Marian Scott** – Analytics for a digital earth addressing climate and environmental change
- 5.30-7pm: Drinks & nibbles reception at the [Wardlaw Museum](#) (2-minute walk from St Salvator's Quad)
- 7.30pm-: Workshop dinner at the [Doll's House](#)

Tuesday, 7th June

- **Session 2, Part 2: Land, Monsoons and Precipitation** (Chair: Spencer Hill)
 - 9-9.30am **Ruth Geen** – Land evaporation biases link to East Asian rainfall shifts across AMIP simulations
 - 9.30-10am **Marianne Pietschnig** – The influence of vegetation and soil moisture on continental precipitation change in the tropics
- 10-10.30am: DISCUSSION on Session 2 (Moderator: Cathy Hohenegger)
- 10.30-11am: COFFEE BREAK (Lower College Hall)
- **Session 3, Part 1: The Land Surface and Hydrology** (Chair: Alexis Berg)
 - 11-11.30am **Steve Sherwood** – What is aridity?
 - 11.30-12pm **Lucas Vargas Zeppetello** – Soil Moisture Trends: Can simple models tell us how soil moisture is evolving across the land surface?
 - 12-12.30pm **Alistair Jump** – Progress in understanding and predicting the where and the why of beech forest decline across Europe
- 12.30-1.30pm: LUNCH (Lower College Hall)
- **Session 3, Part 2: The Land Surface and Hydrology** (Chair: Vincent Humphrey)
 - 1.30-2pm **Alexis Berg** – Asymmetry in the continental water cycle response to warming for cold and warm climates
 - 2-2.30pm **Justin Mankin** – Plants and the future of continental aridity
 - 2.30-3pm **Jack Scheff** – Why do the global warming responses of land-surface models and climatic dryness metrics disagree?
- 3-3.30pm: COFFEE BREAK (Lower College Hall)
- **Session 3, Part 3: The Land Surface and Hydrology** (Chair: Justin Mankin)
 - 3.30-4pm **Kaighin McColl** – A simple theory of continental evapotranspiration
 - 4-4.30pm **Vincent Humphrey** – How droughts impact terrestrial carbon uptake in Earth system models
 - 4.30-5pm **Alex Konings** – Evapotranspiration frequently increases during droughts (*virtual presentation*)
- 5-5.30pm: DISCUSSION on Session 3 (Moderator: Marysa Laguë)
- **6pm: Public Lecture by Prof. Steve Sherwood** – ‘Climate Change & Climate Enigmas’ (School 2)
- 7-8pm: Drinks reception following public lecture (Upper College Hall)

Wednesday, 8th June

- **Session 4, Part 1: Extreme Events over Land** (Chair: Angeline Pendergrass)
 - 9-9.30am **Karen McKinnon** – Hot extremes have become drier in semiarid regions: a case study of the US Southwest (*virtual presentation TBC*)
 - 9.30-10am **Mike Byrne** – Amplified warming of extreme temperatures over tropical land
 - 10-10.30am **Yi Zhang** – An upper bound for extreme temperatures over midlatitude land
- 10.30-11am: COFFEE BREAK (Lower College Hall)

- **Session 4, Part 2: Extreme Events over Land** (Chair: Yi Zhang)
 - 11-11.30am **Aiguo Dai** – Thermodynamic changes over land and their link to precipitation response to global warming
 - 11.30-12pm **Spencer Hill** – Interannual variability in seasonal-mean and daily-extreme rainfall in the Indian summer monsoon
- 12-12.30pm: DISCUSSION on Session 4 (Moderator: Lucas Vargas Zeppetello)
- 12.30-1.30pm: LUNCH (Lower College Hall)
- **Session 5: Computational Techniques & Modelling** (Chair: Andrew Schurer)
 - 1.30-2pm **Manoj Joshi** – Predicting the surface urban heat island using climatic and land surface information (*virtual presentation*)
 - 2-2.30pm **Angeline Pendergrass** – How do regional distributions of daily precipitation change under warming?
 - 2.30-3pm **David Sexton** – Towards including process-based metrics in regional climate projections
- 3-3.30pm: COFFEE BREAK (Lower College Hall)
- **Session 6: Historical Climate Change** (Chair: David Sexton)
 - 3.30-4pm **Gabi Hegerl** – Observational constraints on future climate risk
 - 4-4.30pm **Flavio Lehner** – Alternative historical realizations of tropical sea surface temperature and their influence on continental climate
 - 4.30-5pm **Andrew Schurer** – Role of forcing and three major modes on decadal climate variability and change
- 5-5.30pm: DISCUSSION on Sessions 5 and 6 (Moderator: Michela Biasutti)
- 5.30-6pm: Takeaways from workshop overall (Byrne, Hegerl, Scheff to lead)
- DEPART!